RAJ BHARAT RASAL

Backend & Distributed Systems Architect

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PROFESSIONAL SUMMARY

Undergraduate Computer Science Engineer specializing in backend systems architecture and distributed computing. Proven track record designing zero-trust protocols, dynamic tensor-sharding networks (SATURDAY/SUNDAY), and high-throughput FastAPI microservices. Passionate about verifiable computation, fault-tolerant orchestration, and scalable low-latency infrastructure.

TECHNICAL & ARCHITECTURAL SKILLS

Languages: Python, C++, SQL (PostgreSQL, SQLite)

Distributed Systems & Architecture: Distributed Systems Architecture, Tensor Sharding & Compute Networks, Decoupled Orchestration, Zero-Trust Communication Protocols, Proof-of-Compute Verification, System Design & Scalability

Backend & Cloud: FastAPI, RESTful API Design, Asynchronous Programming (asyncio), Tailscale VPN & Mesh Networking, Redis Streams & Message Passing

Security & Systems: Vulnerability Research & Bug Bounty, Sandboxing & Isolation (Firecracker)

Soft Skills: Technical Architecture Documentation, Conceptual Simplification

FEATURED SYSTEMS & PROJECTS

SATURDAY / SUNDAY — Architect & Lead Developer

Jan 2026 - Present

Tech Stack: Python, C++, Tensor Sharding, Zero-Trust Protocols, LAN, Tailscale VPN, Socket Programming

- Built a distributed compute network to run large local LLMs, solving hardware limitations of processing complex AI models on single devices.
- Implemented automatic hardware discovery and dynamic tensor sharding to split a single model's processing across heterogeneous Mac and Windows nodes via LAN and Tailscale VPN mesh.
- Designed a zero-trust communication protocol between nodes, utilizing proof-of-compute verification to detect and reject nodes returning corrupted or falsified matrix calculation results.
- **Key Architecture:** Decoupled coordinator-worker model with heartbeats; distributed matrix multiplication with Tailscale mesh tunnels; sub-millisecond node health checks; Byzantine fault tolerance through spot verification.

F.R.I.D.A.Y — Multi-Agent Communication Backend — Lead Backend Architect & Developer

Jun 2026 - Present

Tech Stack: Python, FastAPI, Baileys API (WhatsApp), APScheduler, asyncio, LLM Intent Routing

- Originated at SummerHacks 2026 in a team of 4; currently driving continuous backend evolution and multi-agent infrastructure.
- Designed a strictly decoupled 3-agent architecture (Herald, Nexus, Dispatch) enforcing complete separation of concerns so agents never call each other directly, eliminating circular deadlocks.
- Implemented LLM intent classification with confidence-based routing logic to catch ambiguous queries and trigger clarification flows before dispatching execution tasks.
- Engineered resilient background job scheduling and automated task pipelines with robust fault recovery.
- **Key Architecture:** 3-tier mediator topology; zero direct agent coupling; confidence threshold routing (<0.85 triggers conversational disambiguation); asynchronous message queue via FastAPI.

SecureVerse — Decentralized Zero-Trust Architecture — Systems Architect

Oct 2025 - Dec 2025

Tech Stack: Python, Cryptography, Zero-Trust Architecture, Distributed Ledger, REST APIs

- Architected a decentralized identity and zero-trust verification platform engineered to prevent credential theft and unauthorized access in enterprise networks.
- Formulated the end-to-end cryptographic trust chain, policy evaluation engine, and auditable logging system.
- Awarded 3rd Place at the Deal Dome Competition for superior architectural soundness and threat mitigation design.
- **Key Architecture:** Mutual TLS, cryptographic assertion validation, state machine based authorization engine, zero persistent plain text secrets.

EXPERIENCE & LEADERSHIP

Independent Tutoring Practice — Mathematics & Physics Tutor (Grades 10–12)

Jun 2024 - Present

- Taught 50+ high school students, translating complex scientific, mathematical, and algorithmic concepts into clear, structured lessons.
- Developed diagnostic assessment frameworks to identify conceptual bottlenecks and tailored individual curricula to improve performance by an average of 22%.

- Honed mastery of first-principles thinking, communication clarity, and pedagogical system design.

SYSTEM DESIGN SPECIFICATIONS & WHITEPAPERS

HAMIA Architecture Specification: Hierarchical Adaptive Multi-Model Intelligence Architecture (Architecture Whitepaper, Oct 2026)

- A cloud-native fusion system that orchestrates a federation of specialized frontier models (12+) through an 8-stage pipeline. Emphasizes evidence-centric fusion over naive voting, using external ground-truth verifiers (Firecracker sandboxes, SymPy) and a cyclic adversarial critic to prevent multi-model ensembles from amplifying shared bias.

EDUCATION

ITM Skills University — Bachelor of Technology (B.Tech)

Aug 2025 - May 2029

Field: Computer Science Engineering | Score: Undergraduate (In Progress)

- Specializing in Distributed Systems, Operating Systems, Computer Networks, and AI Systems Design. Active contributor to student engineering and hackathon teams.

Central Board of Secondary Education (CBSE) — Higher Secondary Certificate (12th Grade)

2023 - 2025

Field: Science (PCM) | Score: 83.4%

- Strong foundation in advanced calculus, mechanics, electromagnetic physics, and algorithmic logic.

Central Board of Secondary Education (CBSE) — Secondary School Examination (10th Grade)

2021 - 2023

Field: General Science & Mathematics | Score: 96.0%

- Academic excellence award in Mathematics and Science.

ACHIEVEMENTS & SECURITY RESEARCH

- **3rd Place — Deal Dome Competition:** Awarded 3rd place for designing the architectural specification and threat modeling for SecureVerse, a zero-trust enterprise security framework.
- **Security Research: OPPO Find X3 Pro Vulnerability:** Identified and responsibly reported an Open Redirect vulnerability in the OPPO Find X3 Pro flagship device software ecosystem, acknowledged in HackerOne Hall of Fame.

CERTIFICATIONS & ACCREDITATIONS

- **Advanced Backend Engineering & Distributed Systems Architecture** — Engineering Systems Council (2025)
- **Responsible Disclosure & Security Research Recognition** — HackerOne (2025)